3 day 1020
01/04/2025

- Type of machine learning

- Supervised
  - Regression
  - classification
- unsupervised
  - clustering
  - dimensionality Reduction
  - Anamoly detection
  - Association
- Semi supervised
- Reinforcement

Supervised learning :- both input & output.

Ex:-

iq	cgpa	Y/N placement
87	7.1	Y
111	8.9	Y
75	6.3	N

Supervised learning
- Regression
- classification

for, Data
- Numerical
  - cgpa
  - IQ
  - Age
  - wt
- categorical
  - gender
  - Nation.

Regression

IQ	capa	Package
87	8.9	6
111	8.9	4

→ Regression.

classification

	capa	
87	8.9	yes
111	8.9	NO

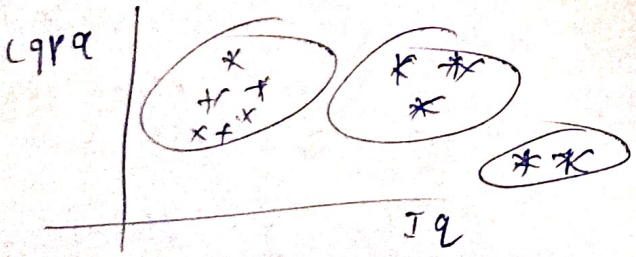
→ classification.

Unsupervised Learning:- only Input

- clustering
- dimensionly Red
- Anomaly detection
- Association Rule learning

Ex:-

IQ	capa	out
70	7.0	?
60	6.0	
100	9.	



IQ capa ↑
IQ ↓ capa ↑
capa ↓ IQ ↑

dimensionally:- Basically an large amount of input data convert into small

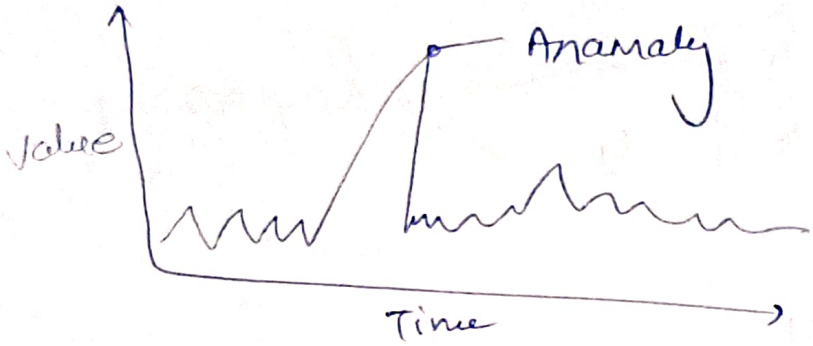
Ex:- 100 input term

or, if like Housing example here

Home → | No of Room | No of Washroom | → convert → square foot

Anamoly detection → Ex- detection in loan approval, credit card processing

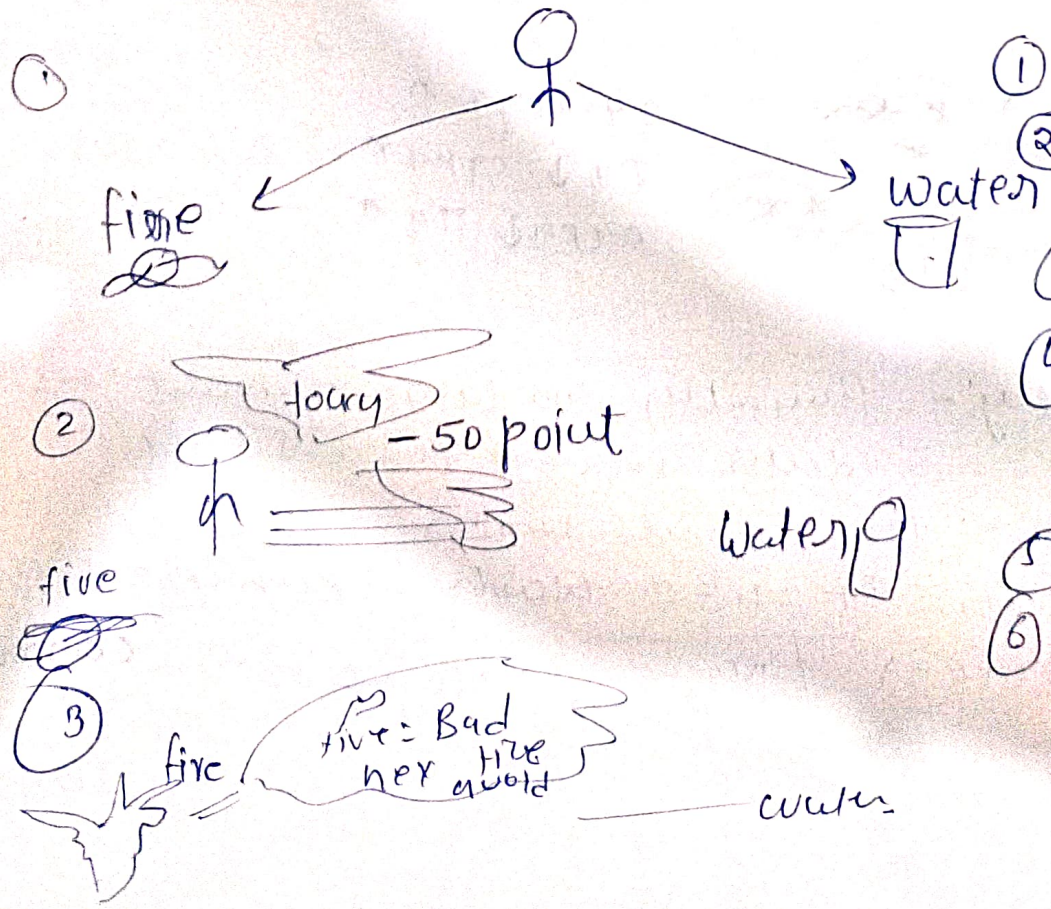
Ex:-



Association-Rule Based :- Ex:- if any person buy an milk so may be they buy bread.

Semi-Supervised:- partially both..

Reinforcement:-



1. observed
2. select Again using policy
3. Action
4. get reward or penalty
5. update policy
6. iterate an optimal policy.